NTUST course: Computer Vision and Applications (CI5336701, 2026 Spring)

Midterm Project: Re-project and mix colors for a 3D model from multiple-view images

Date Due：2026. Apr. 24th, Friday, PM11:59 (~2.5 weeks)

Description

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1. The midterm project is to "Colorize" a 3D model (stored as Santa.xyz) by multiple photos from different views (different intrinsic and extrinsic factors) the scenario of 3D scanning device based on a known structure.
2. The given data are "a 3D model w/o color" and "7 images". Please write a program to parse *.xyz (which is a 3D raw data, each raw represents a 3D vertex and its normal vector). 7 images were taken by two cameras (iphone and SONY DSLR). All images have different K and [R|t].
3. In this assignment, you do NOT need to solve individual **K** and [**R**|**t**]. Instead, to have extremely correct color alignment, you need to estimate "projection matrixes" (the combination of **K** and [**R**|**t**]) carefully. All features were selected manually (No need to develop auto-detection solution, and No need of GUI for picking 2D / 3D points). And try to project images onto the 3D model and consider the visibility and occlusion. For a vertex which can be seen from two or more views, you need to blend or mix color to make it look consistent.
4. Deliverable: There are three types of data you should provide:
   1. Result: color xyz file (or ply file): each row shows x,y,z,R,G,B and A data
   2. Source code in python, Matlab, or C++/C etc. with simple comment: A program is able to import all images and output Santa.xyz (having correct size).
   3. Two-page report (in English) saved in PDF format: to describe what points you select, and how you solve it.
   4. (Optional) Execution file (.exe) for this project, if written in C/C++, please compile into exe. For python / matlab user, you don't need to generate exe files. (Compulsory for C/C++, but Optional for python/matlab).

   Please zip all your files, then, upload on moodle2 by due date.
5. Reference score

   70%: Completeness: Correctly estimate all projection matrixes and creating a COLOR 3D Santa.xyz model.

   30%: Correctness and performance: Alignment accuracy, consistent color, few visual defects, etc.

Hint:

1. There are several ways you can improve the quality: a) Carefully select the corresponding 3D / 2D points, b) Avoid degenerated cases, c) Evaluate re-projection errors, d) Mix color according to view-dependent attenuation factors, e) Reject back-faces (here is “back”-points) who are invisible, f) Consider the distance effect to reject “occluded” points
2. This midterm project will be **15%** of final grade.
3. You can use Meshlab software to assist you for query the 3D coordinate of every vertex in a XYZ file.

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